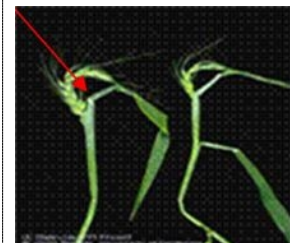


Russian wheat aphid on wheat

Diuraphis noxia Yesinde Kish Kish in Amharic language



Aphid feeds on wheat leaf, 2 mm long (J. Kelly Clark)



Curled flag leaf from wheat aphid infestation (J. Kelly Clark)



White streaks on wheat leaf caused by aphid feeding (Koishibaev, M) 1

Prevention	Monitoring	Direct Control	Direct Control	Restrictions
- Remove wheat residue from the previous season before planting. However, no major insect pests diseases had been in the previous cereal, then leave residues as they improve the soil, and reduce water evaporation. - Plant early in the rainy season - Seed treatment (Apron Star & Thiamethoxam) (see under yellow direct control) - Remove alternative hosts (barley, triticale, rye, oats and wild grain species) - Rotate with pulse crops and large cereals (e.g. Maize & Sorghum)	- Scout fields regularly following seedling emergence (especially during dry period) - Check a minimum of 10 random plants across the field. - Look for gray to light green soft-bodied tiny insects. Aphids are only 1 to 2 mm long - At heavy infestation, stunted plant, curled leaves, white/yellow colour stripes on the leaves - Consider action if the 2-5 aphids/ plant are found at 2 leaf-early tillering stage, 10-12 aphids/ plant at late tillering-first node stage, 20-22 at booting, and 30-32 at head exertion stage or later	- Spray homemade garlic (onion) mixture to repel aphids - Spray homemade neem leaf extract to control aphid - Dilute 1 liter of neem leaf extract with 9 liters of water; Add 100 ml of soap; Stir well spray on the infested plants. - Spray Chili extracts. In a pot, boil ripe pods or chili seeds in water for 15-20 minutes. - Take the pot from the fire and add 3 liters of water. Cool and strain. Add soap. Stir well & spray on infested plants	- Seed treatments with systemic insecticide help to some extent to prevent aphid attack, but only on younger plants. Seed treatment must be done by trained, certified service providers. It cannot be done by the farmer as treatment method too dangerous. Buy treated seeds on advise of local extensionist. - When using a pesticide or botanical, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, pre-harvest interval, max number of sprays, restricted re-entry interval. Do not empty into drains and water sources. - Always consult recent list of registered pesticides in Ethiopian Ministry of Agriculture, Animal and Plant Health Regulatory Directorate. - Consider spraying products based on Thiamethoxam 20% + metalaxyl - 20% + difenoconazole 2% (Apron Star 42 WS, and others), Systemic in seeds and young plant4, Metalaxyl & difenoconazole are fungicides. Use 250 gramm Apron Star 42 WS /100kg seed, but check the product label & follow instructions. - Thiamethoxam – based products; Neonictinoid, Systemic (IRAC, 4A)	- Thiamethoxam is not WHO-classified , but considered slightly acute hazardous in normal use. Broad spectrum insecticide with systemic action. Metalaxyl is WHO toxicity class III (slightly acute hazardous), controls fungal diseases. Difenoconazole: WHO class III (slightly Hazardous), broad spectrum fungicide. Both fungicides do not help to control aphids, but are often mixed with insecticide products - WHO Class III (Moderately hazardous), It is registered for seed treatment.



Ethiopia

CREATED/UPDATED: May 2015

AUTHOR(S): Frehiwot Sileshi/Ambo University, Ethiopia, phone no., email: frehiwotsileshi19@gmail.com

EDITED BY: Plantwise

LOSE LESS, FEED MORE

PlantwisePlus is a CABI-led global initiative www.plantwise.org
©CAB International. Published under a CC BY-NC-SA 4.0 licence